Lines, Triangles & Polygons

Section 1: Coordinate Geometry of Lines

1 Find the perpendicular distance from the point $(3, -2)$ to the line $5x - 12y + 8 = 0$.

a. 2.5

b. 3.6

c. 4.0

d. 4.5

2 What is the area of the triangle with vertices at $(0, 0)$, $(6, 2)$, and $(4, 8)$ in the Cartesian plane?

a. 16

b. 20

c. 24

d. 30

3 What is the acute angle of intersection between the lines $y = 2x + 1$ and $y = -3x + 4$?

a. 45°

b. 60°

c. 30°

d. 75°

Section 2: Triangle Theorems & Inequalities

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4 A triangle has side lengths of 7 and 11. Which of the following CANNOT be the length of the third side?

a. 4

b. 5

c. 10

d. 17

5 In $\triangle ABC$, $AB = 8, BC = 6,$ and angle $B = 90^\circ$. A line parallel to BC intersects AB at D and AC at E. If $AD = 6$, what is the length of segment DE?

a. 4.0

b. 4.5

c. 5.0

d. 5.5

6 The side lengths of a triangle are $5, 12,$ and 13. What is the radius of the incircle (inradius)?

a. 1

b. 2

c. 3

d. 4

Section 3: Polygons & Interior/Exterior Angles

7 If the interior angle of a regular polygon is 150°, how many sides does the polygon have?

a. 10

b. 12

c. 15

d. 18

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8 How many diagonals does a regular decagon (10-sided polygon) have?

a. 25

b. 35

c. 45

d. 70

9 What is the area of a regular hexagon with side length 4 cm?

a. $24\sqrt{3}$ cm^2

b. $36\sqrt{3}$ cm^2

c. 48 cm^2

d. $16\sqrt{3}$ cm^2

Section 4: Geometric Transformations & Dilations

10 A triangle with an area of 14 cm^2 is dilated by a scale factor of $k = 3$. What is the area of the image triangle?

a. 42 cm^2

b. 84 cm^2

c. 126 cm^2

d. 168 cm^2

11 What is the matrix representation of a counterclockwise rotation of 90° about the origin?

a. $[[0, -1], [1, 0]]$

b. $[[0, 1], [-1, 0]]$

c. $[[1, 0], [0, -1]]$

d. $[[-1, 0], [0, 1]]$

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12 A line $3x - 4y = 7$ is translated 2 units right and 1 unit up. What is the equation of the new line?

a. $3x - 4y = 9$

b. $3x - 4y = 5$

c. $3x - 4y = 7$

d. $3x - 4y = 11$

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